# Smith chart

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# Smith Chart (Graphical way of finding TL parameters)

- General expression for input impedance is $Z_s = R_0 \dfrac{1 + |K|\angle\phi - 2\beta s}{1 - |K|\angle\phi - 2\beta s}$
- Normalized input impedance is

$$\frac{Z_s}{R_0} = \frac{1 + |K|\angle\phi - 2\beta s}{1 - |K|\angle\phi - 2\beta s}$$

$\dfrac{Z_s}{R_0} = r_a + jx_a$ → Normalized reactance

→ Normalized resistance

But $|k|\angle\phi - 2\beta s = u + jv$ (a complex number)

$$\therefore \quad r_a + jx_a = \frac{1 + u + jv}{1 - u - jv} \times \frac{1 - u + jv}{1 - u + jv} = \frac{1 - u^2 - v^2 + j2v}{(1-u)^2 + v^2}$$

$$\Rightarrow r_a = \frac{1 - u^2 - v^2}{(1-u)^2 + v^2} \quad \text{and} \quad x_a = \frac{2v}{(1-u)^2 + v^2}$$

$$\Rightarrow r_a = \frac{1-u^2-v^2}{(1-u)^2+v^2} \quad \text{and} \quad x_a = \frac{2v}{(1-u)^2+v^2} \rightarrow (2)$$

$\hookrightarrow (1)$

From (1) $\quad \{(1-u)^2+v^2\} r_a - 1 + u^2 + v^2 = 0$

$$r_a - 2ur_a + u^2 r_a + v^2 r_a - 1 + u^2 + v^2 = 0$$

$$u^2(r_a+1) - 2ur_a + v^2(r_a+1) = 1 - r_a$$

$$u^2 - 2u\frac{r_a}{r_a+1} + v^2 = \frac{1-r_a}{1+r_a}$$

$$u^2 - 2u\frac{r_a}{r_a+1} + \frac{r_a^2}{(r_a+1)^2} + v^2 = \frac{1-r_a}{1+r_a} + \frac{r_a^2}{(1+r_a)^2}$$

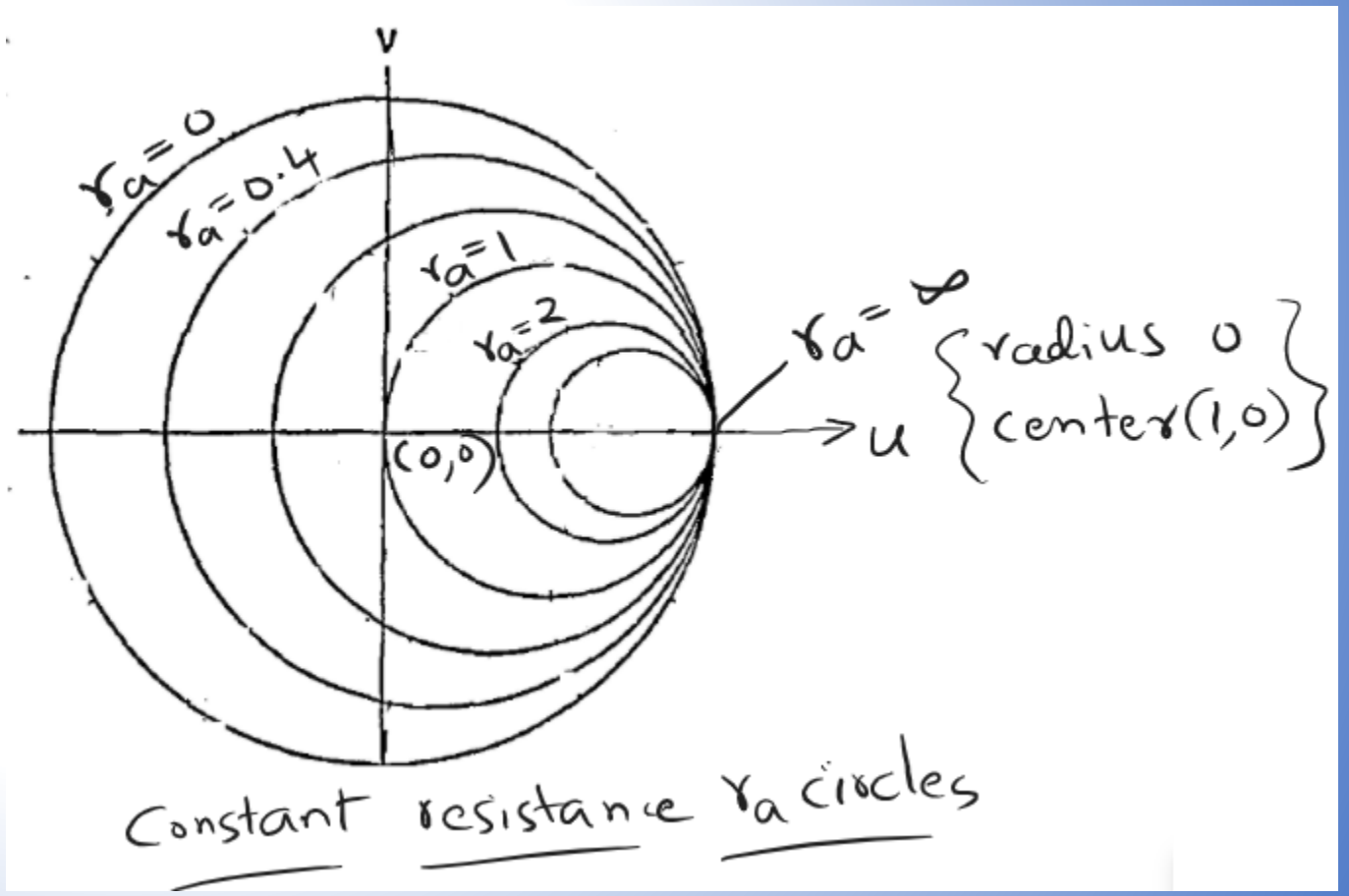
$$\boxed{\left(u - \frac{r_a}{r_a+1}\right)^2 + (v-0)^2 = \frac{1}{(r_a+1)^2}}$$

Similar to equation of a circle $\boxed{(x-g)^2 + (y-h)^2 = r^2}$

$$\left[U - \left(\frac{r_a}{r_a + 1}\right)\right]^2 + V^2 = \frac{1}{(r_a + 1)^2}$$

Center $\left(\frac{r_a}{r_a+1}, 0\right)$, Radius $\left(\frac{1}{r_a+1}\right)$

| $r_a$ | Center | Radius |
|---|---|---|
| 0 | (0,0) | 1 |
| 0.2 | (0.1667,0) | 0.833 |
| 0.4 | (0.285,0) | 0.715 |
| .. | .. | .. |
| 1 | (0.5,0) | 0.5 |
| 2 | (0.667,0) | 0.33 |
| .. | | |
| Infinity | (1,0) | 0 |



$$\Rightarrow r_a = \frac{1-u^2-v^2}{(1-u)^2+v^2} \quad \text{and} \quad x_a = \frac{2v}{(1-u)^2+v^2} \rightarrow (2)$$

$\lfloor (1)$

From (2)

$$(1-u)^2 + v^2 - \frac{2v_a}{x_a} = 0$$

$$(1-u)^2 + v^2 - \frac{2v_a}{x_a} + \frac{1}{x_a^2} = \frac{1}{x_a^2}$$

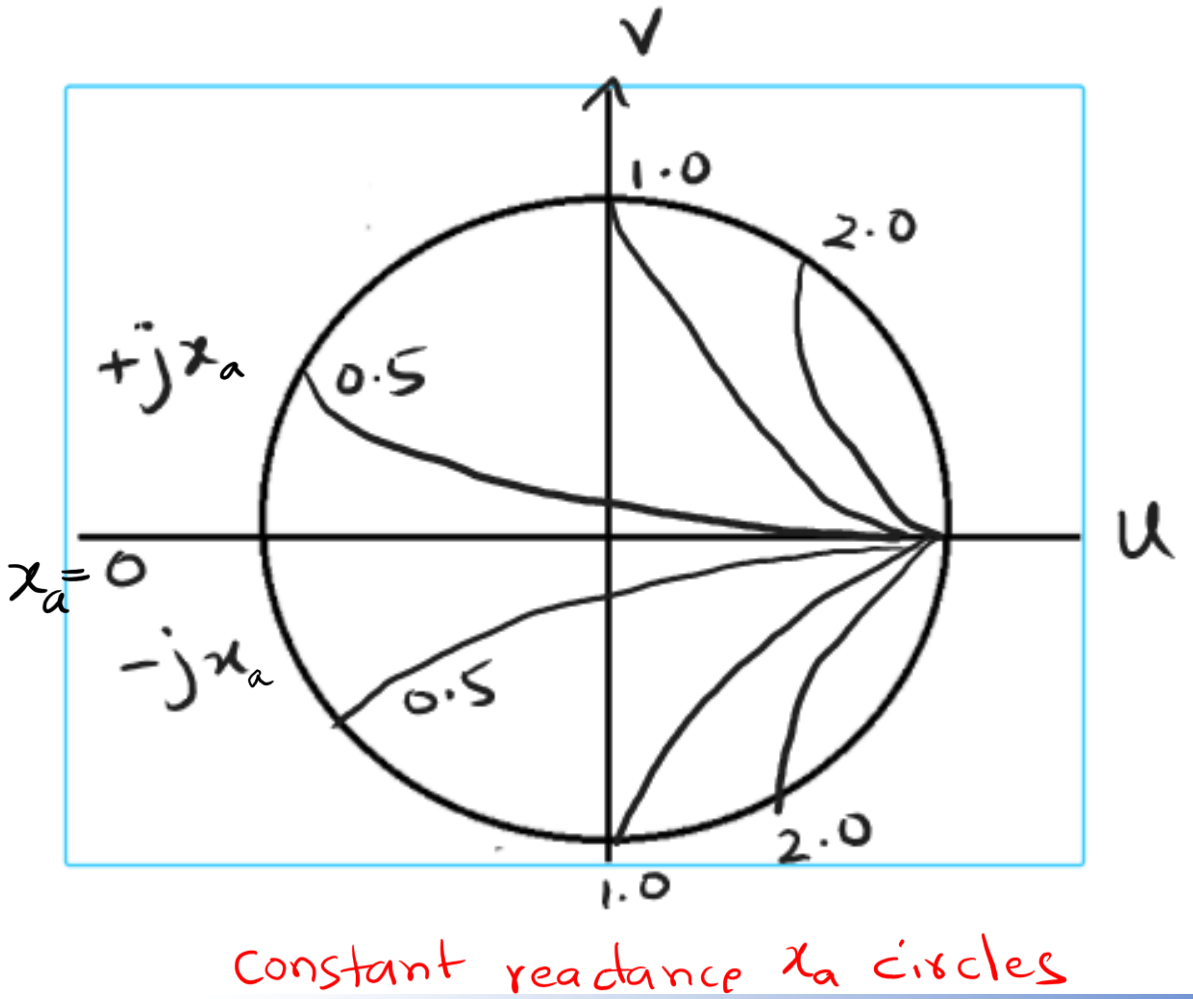
$$(u-1)^2 + \left(v - \frac{1}{x_a}\right)^2 = \frac{1}{x_a^2}$$

Center $\left(1, \frac{1}{x_a}\right)$ ; radius $= \frac{1}{x_a}$

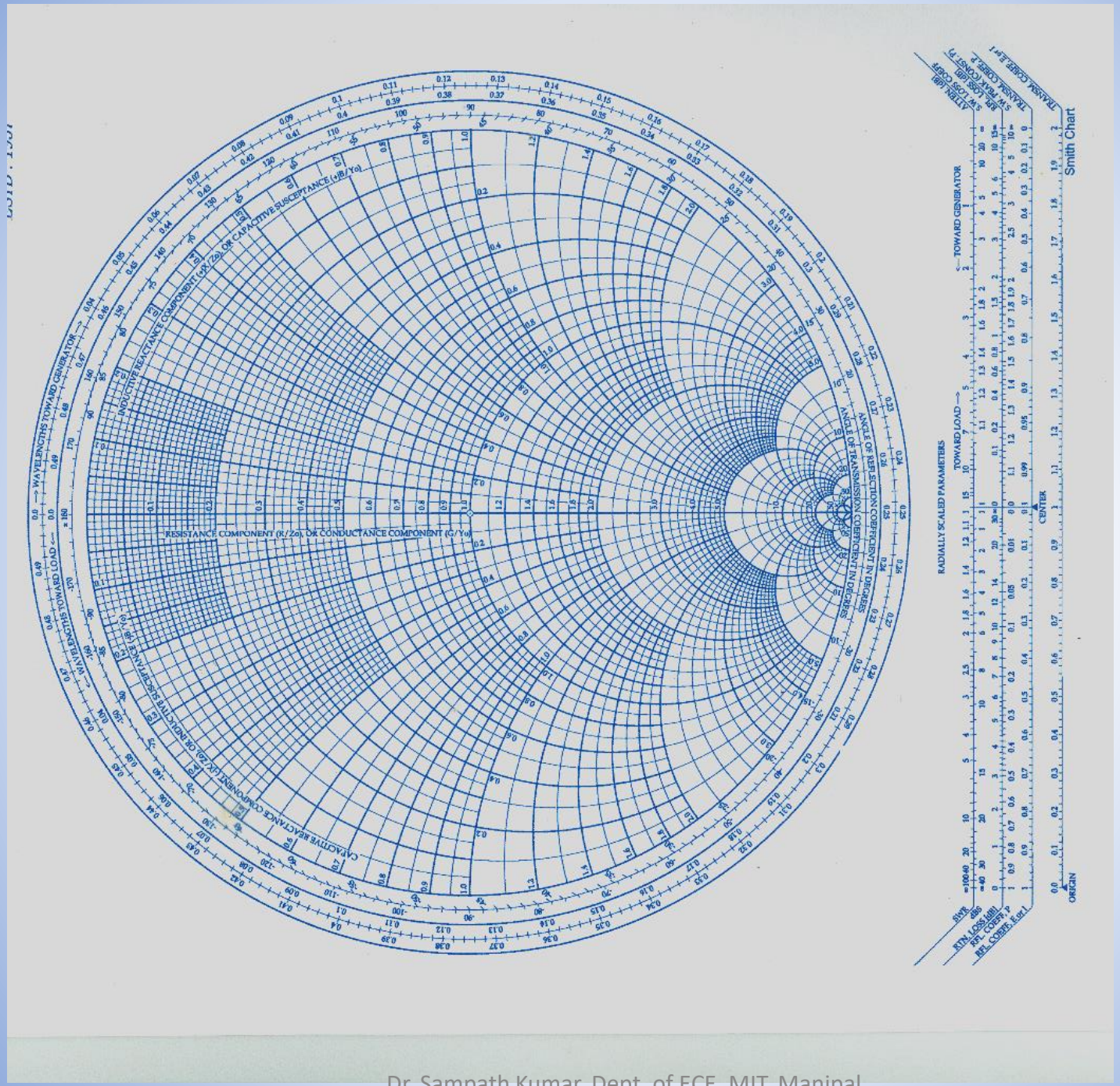
Center $\left(1, \frac{1}{x_a}\right)$ ; radius $= \frac{1}{x_a}$

| $x_a$ | center | radius |
|---|---|---|
| 0 | $(1, \infty)$ | $\infty$ |
| 0.2 | $(1, 5)$ | 5 |
| 0.4 | $(1, 2.5)$ | 2.5 |
| ⋮ | ⋮ | ⋮ |
| 1 | $(1, 1)$ | 1 |
| 2 | $(1, 0.5)$ | 0.5 |
| ⋮ | ⋮ | ⋮ |
| $\infty$ | $(1, 0)$ | 0 |

[Similarly for $-0.2, -0.4, \ldots -\infty$]



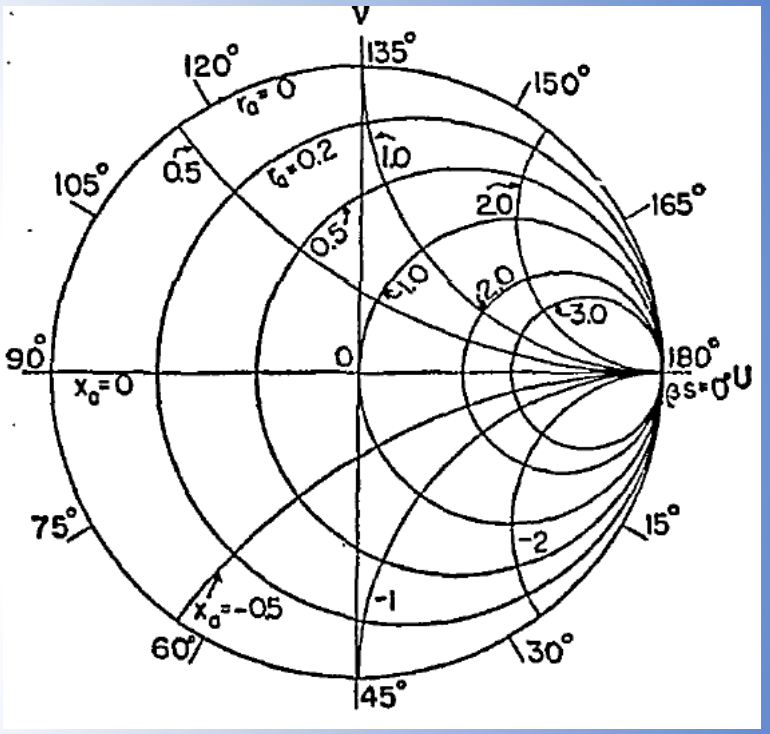

Constant reactance $x_a$ circles

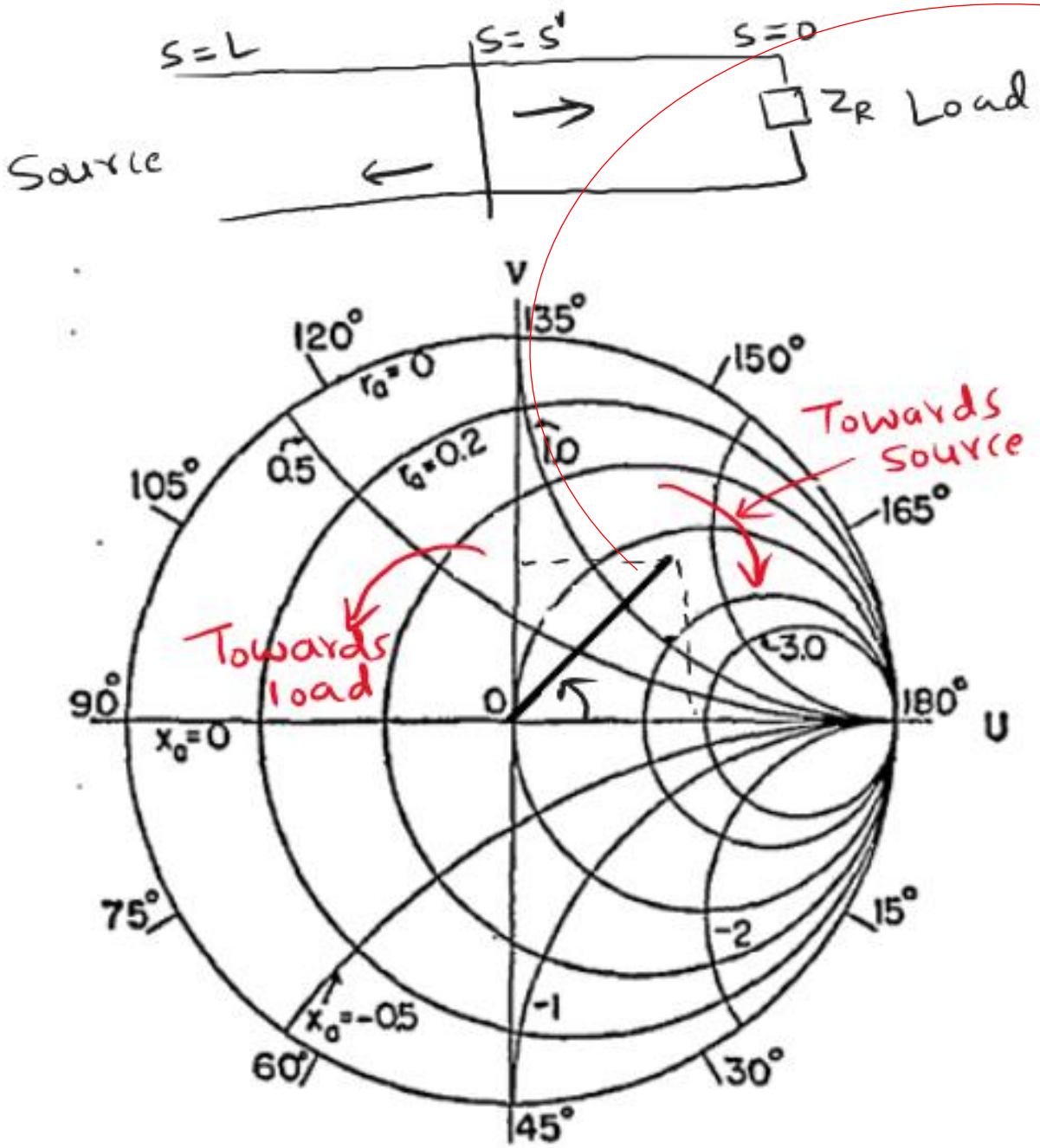

Smith Chart
RESISTANCE COMPONENT (R/Zo), OR CONDUCTANCE COMPONENT (G/Yo)
INDUCTIVE REACTANCE COMPONENT (+jX/Zo), OR CAPACITIVE SUSCEPTANCE (+jB/Yo)
CAPACITIVE REACTANCE COMPONENT (-jX/Zo), OR INDUCTIVE SUSCEPTANCE (-jB/Yo)
WAVELENGTHS TOWARD GENERATOR
WAVELENGTHS TOWARD LOAD
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
ANGLE OF TRANSMISSION COEFFICIENT IN DEGREES
RADIALLY SCALED PARAMETERS
TOWARD GENERATOR
TOWARD LOAD
CENTER
ORIGIN

[Smith chart]

# Characteristics of Smith Chart:

1) The constant $r_a$ and constant $x_a$ loci form two families of orthogonal circles in the chart
2) The constant $r_a$ and constant $x_a$ circles all pass through the point (u=1 & v=0)
3) Upper half of the diagram represents +jx and lower half represents –jx
4) For admittance, the constant $r_a$ circles become constant $g_a$ circles; the constant $x_a$ circle becomes constant susceptance $b_a$ circles
5) The distance around the Smith chart once is one half of the wavelength (λ/2)






$$|k|e^{j(\phi-2\beta s')} = u+jv$$

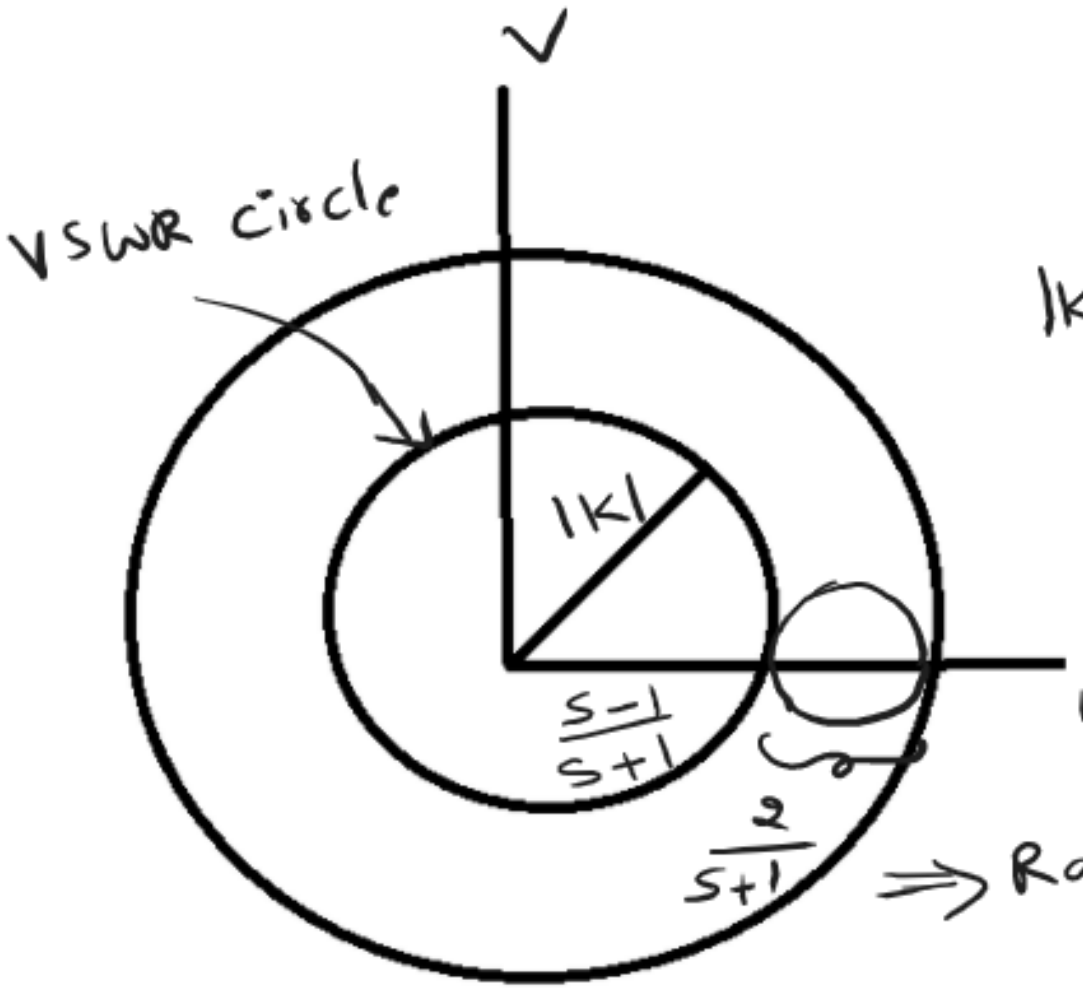
$\phi - 2\beta s' + 2\beta\Delta s$
(Towards load)

$$2\beta\Delta s = 2\pi$$

$$\Delta s = \frac{\pi}{\beta} = \frac{\pi\lambda}{2\pi}$$

$$\Delta s = \frac{\lambda}{2}$$

6) At a point of $Z_{min} = 1/S$, there is $V_{min}$ on the line (Normalized impedance ($Z_{min}/R_0 = 1/S$)
7) At a point of $Z_{max} = S$, there is $V_{max}$ on the line
8) The horizontal radius to the right of chart center corresponds to $V_{max}$, $I_{min}$, $Z_{max}$ and S (SWR)
9) The horizontal radius to the left of chart center corresponds to $V_{min}$, $I_{max}$, $Z_{min}$ and 1/S
10) Since the normalized admittance $Y_a$ is a reciprocal of the normalized impedance Z, the corresponding quantities in the admittance chart are $180^0$ out of phase with those of the impedance chart.




$$|k| = \frac{S-1}{S+1}$$

$$\Rightarrow \text{Radius} = \frac{1}{1+S} = \frac{1}{1+r_a}$$

$$\therefore r_a = S$$

# Example:

- A 50 Ω lossless line has $Z_R = 40 + j30\ \Omega$ is at s = 0. Find K

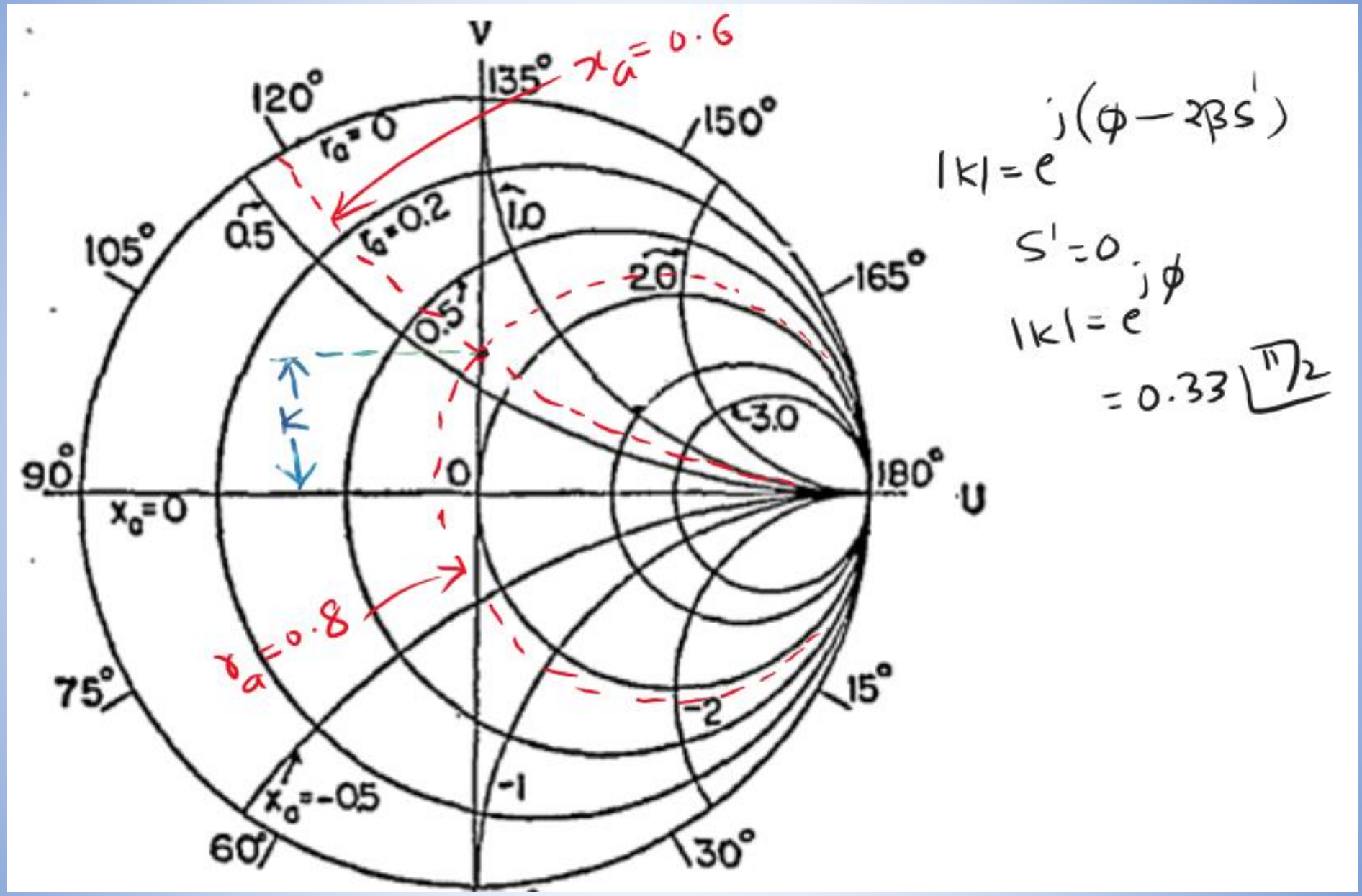
Solution:

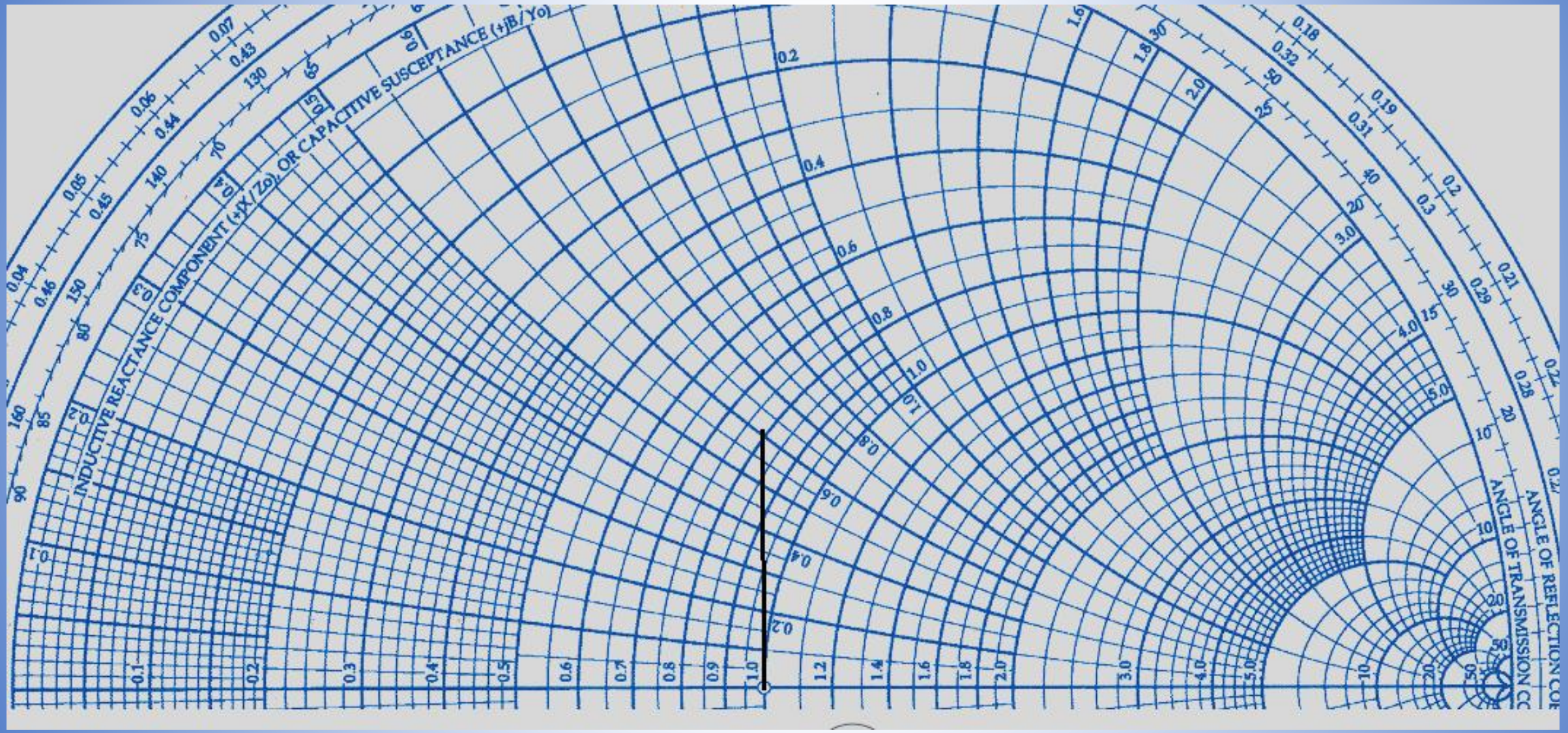
From definition K = $\frac{Z_R - R_0}{Z_R + R_0} = \frac{-10 + j30}{90 + j30} = 0.333 \angle \pi/2$

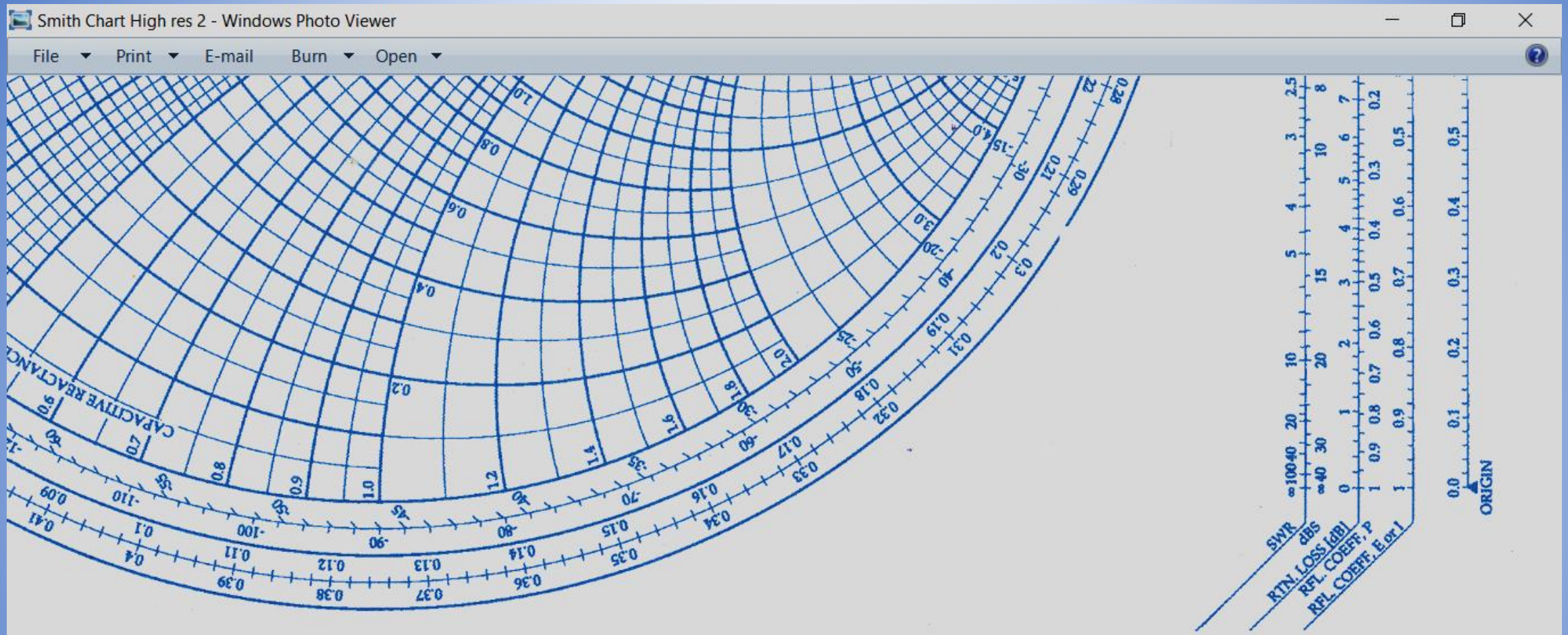
Smith chart: Use normalized impedances

$$\frac{Z_s}{R_0} = r_a + jx_a = \frac{Z_R}{R_0} \text{ at } s = 0$$

$$= \frac{40 + j30}{50} = 0.8 + j0.6$$


V
U
120°
135°
150°
105°
165°
90°
180°
75°
15°
60°
30°
$r_0 = 0$
0.2
0.5
1.0
2.0
3.0
0.5
0
-0.5
-1
-2
$x_0 = 0$
$x_0 = -0.5$
$x_a = 0.6$
$r_a = 0.8$
K
$|k| = e^{j(\phi - 2\beta s')}$
$s' = 0$
$|k| = e^{j\phi}$
$= 0.33 \angle 11/2$


CAPACITIVE SUSCEPTANCE (+jB/Yo)
INDUCTIVE REACTANCE COMPONENT (+jX/Zo)
ANGLE OF TRANSMISSION CO
ANGLE OF REFLECTION CO

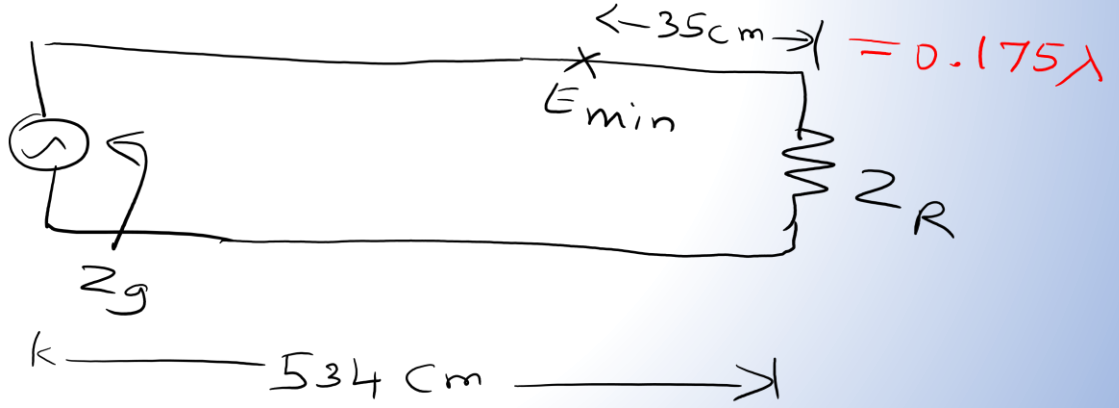



Use divider or compass for magnitude,
Measure angle of reflection coefficient

# Problem 6:

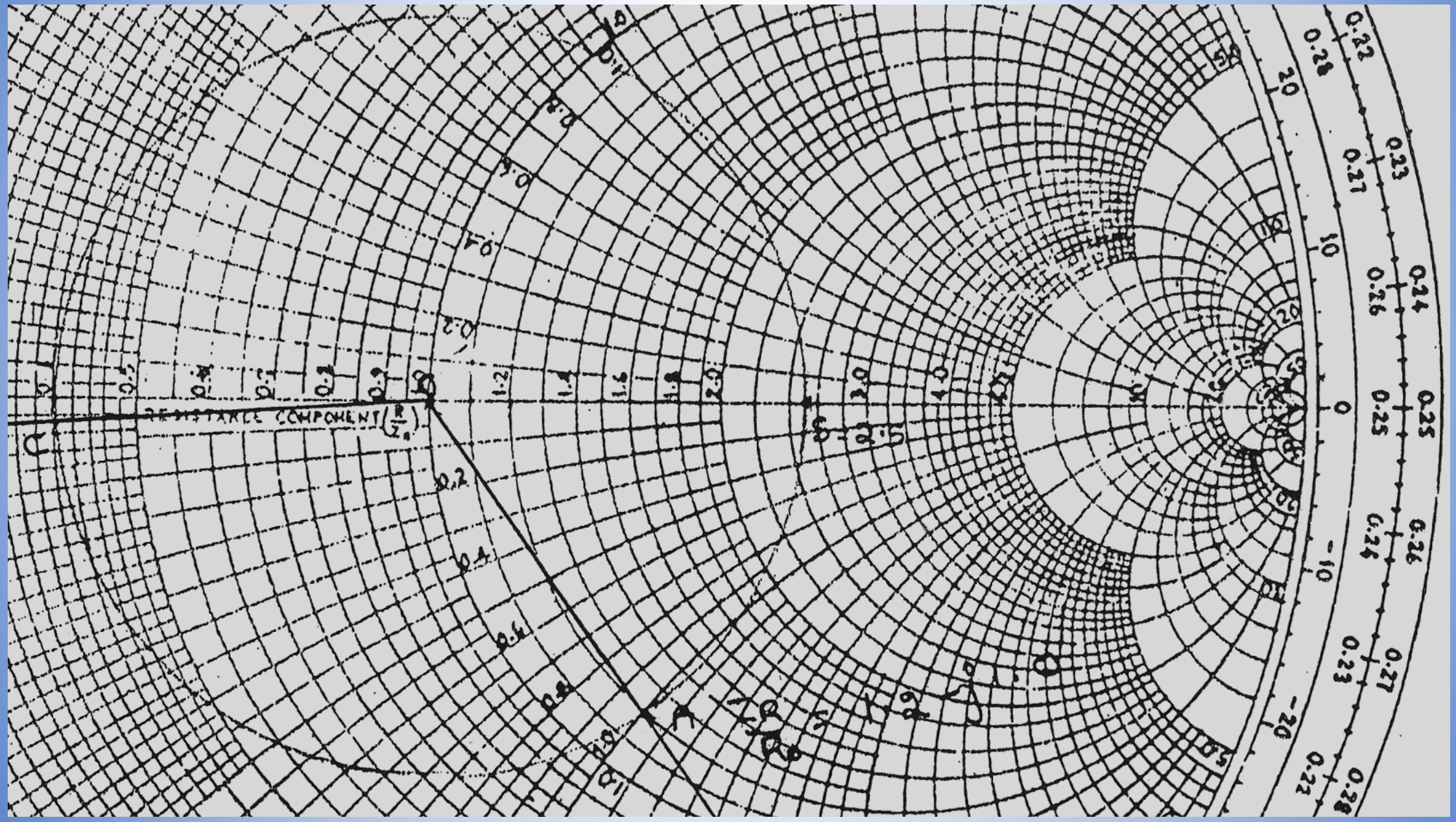
- A transmission line is terminated with an unknown impedance $Z_R$. It is found that the first voltage minimum occurs at a distance of 35 cm from the load. The total length of line is 534 cm. Given that S = 2.5 and $R_0$ = 100 Ω. The operating frequency is 150 MHz. Find $Z_R$ and $Z_g$.

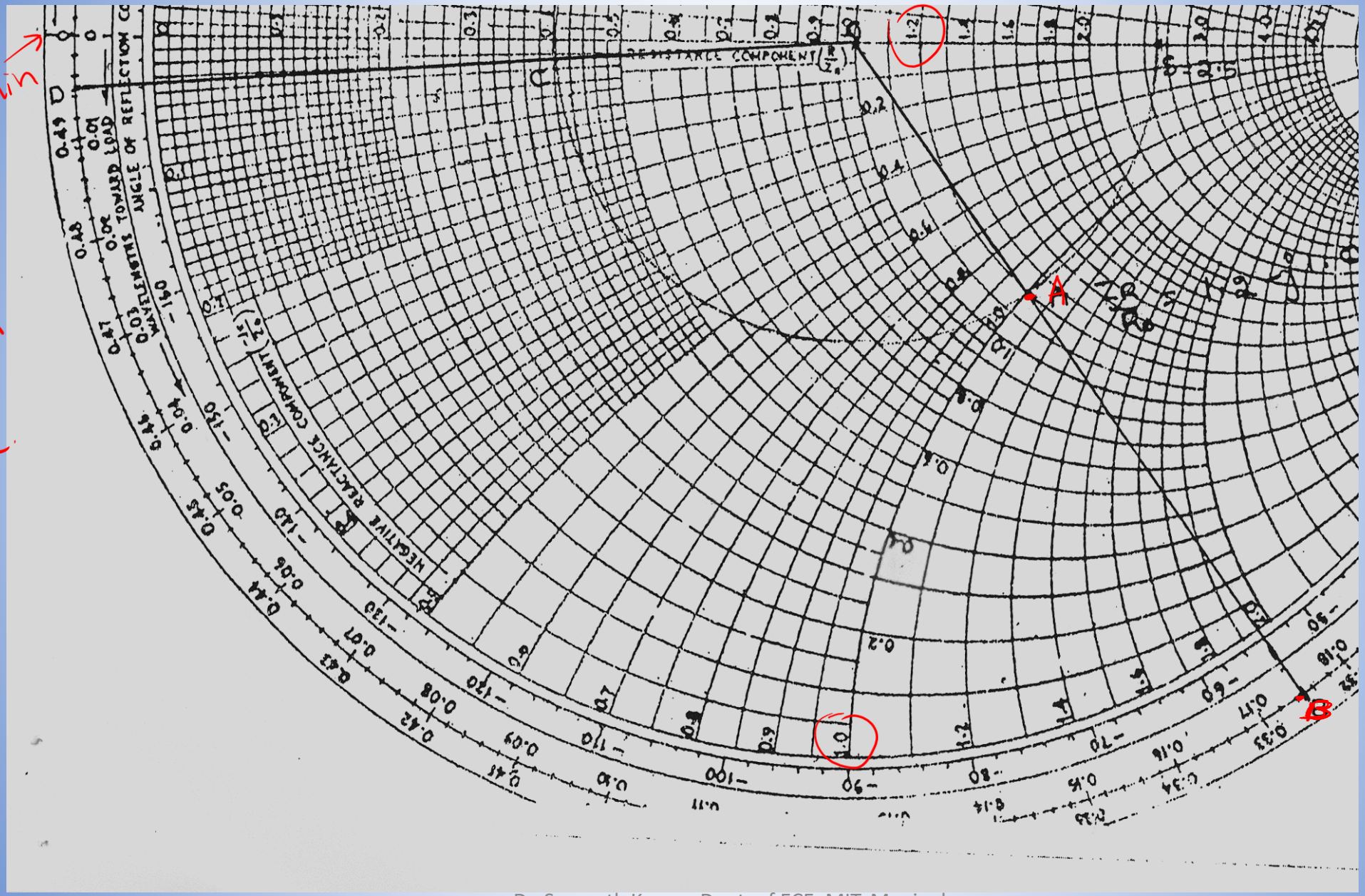
$$\lambda = \frac{c}{f} = \frac{3\times10^8}{150\times10^6} = 200\text{cm}$$

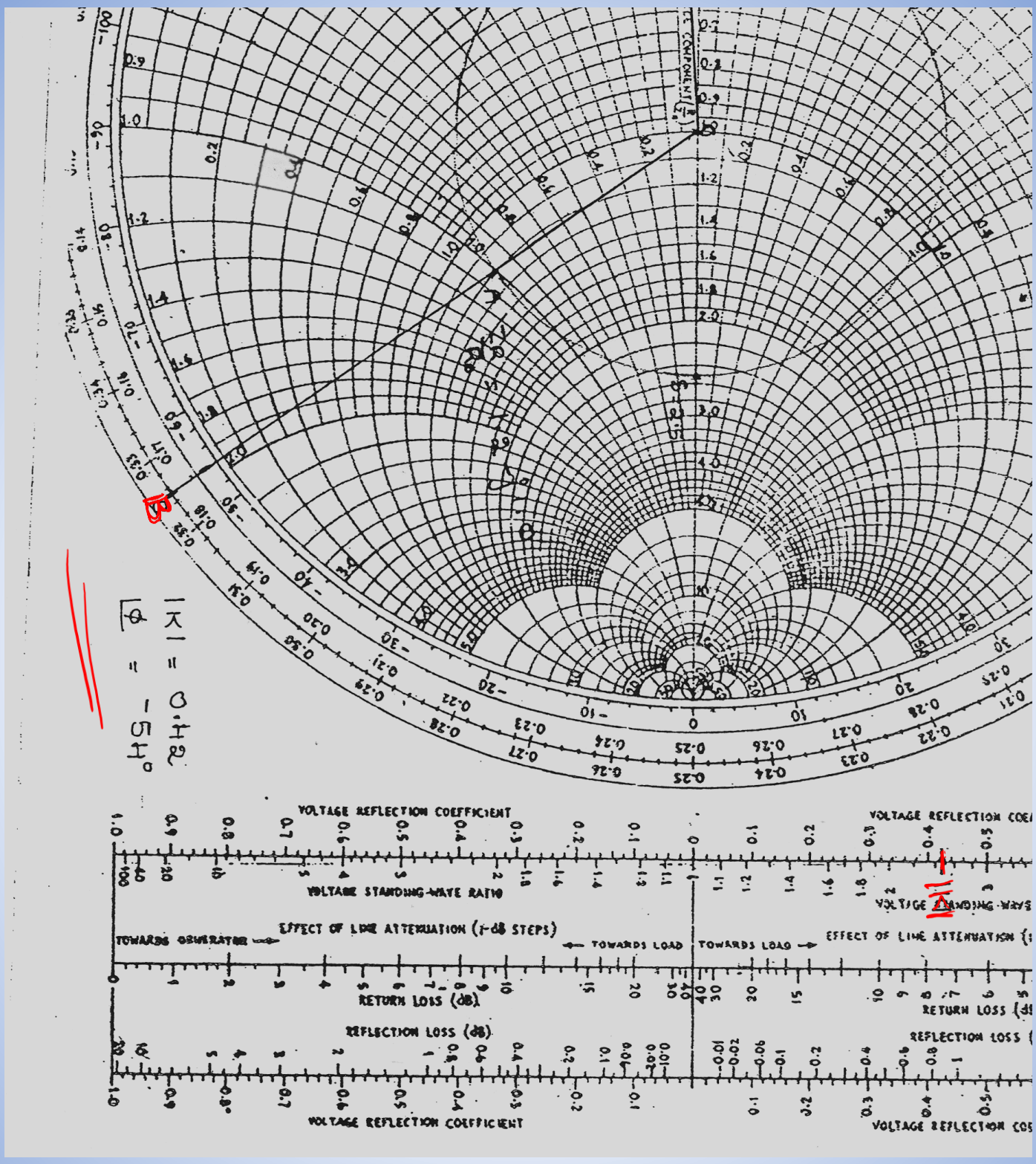


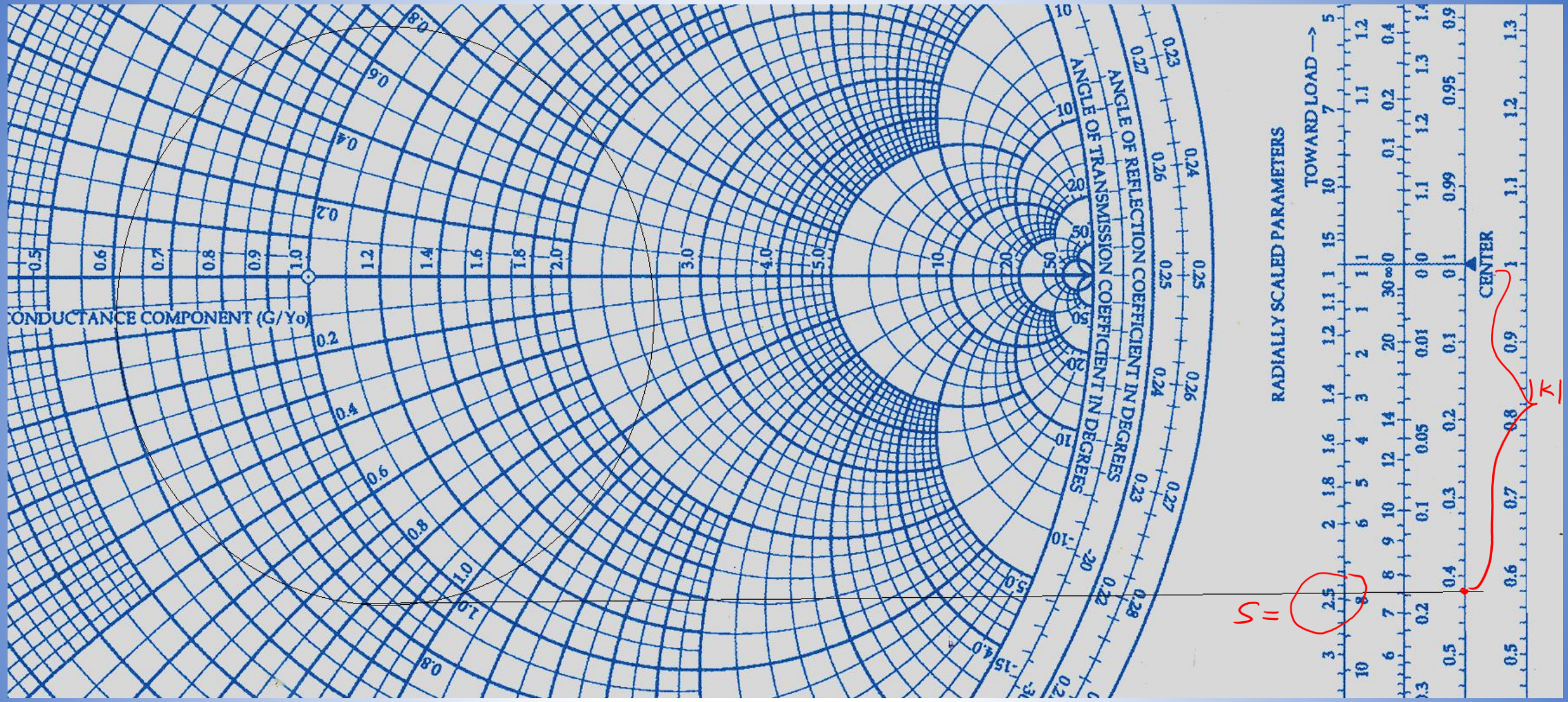

$$2.67 - 0.175 = 2.495 - 2 = 0.495\lambda$$

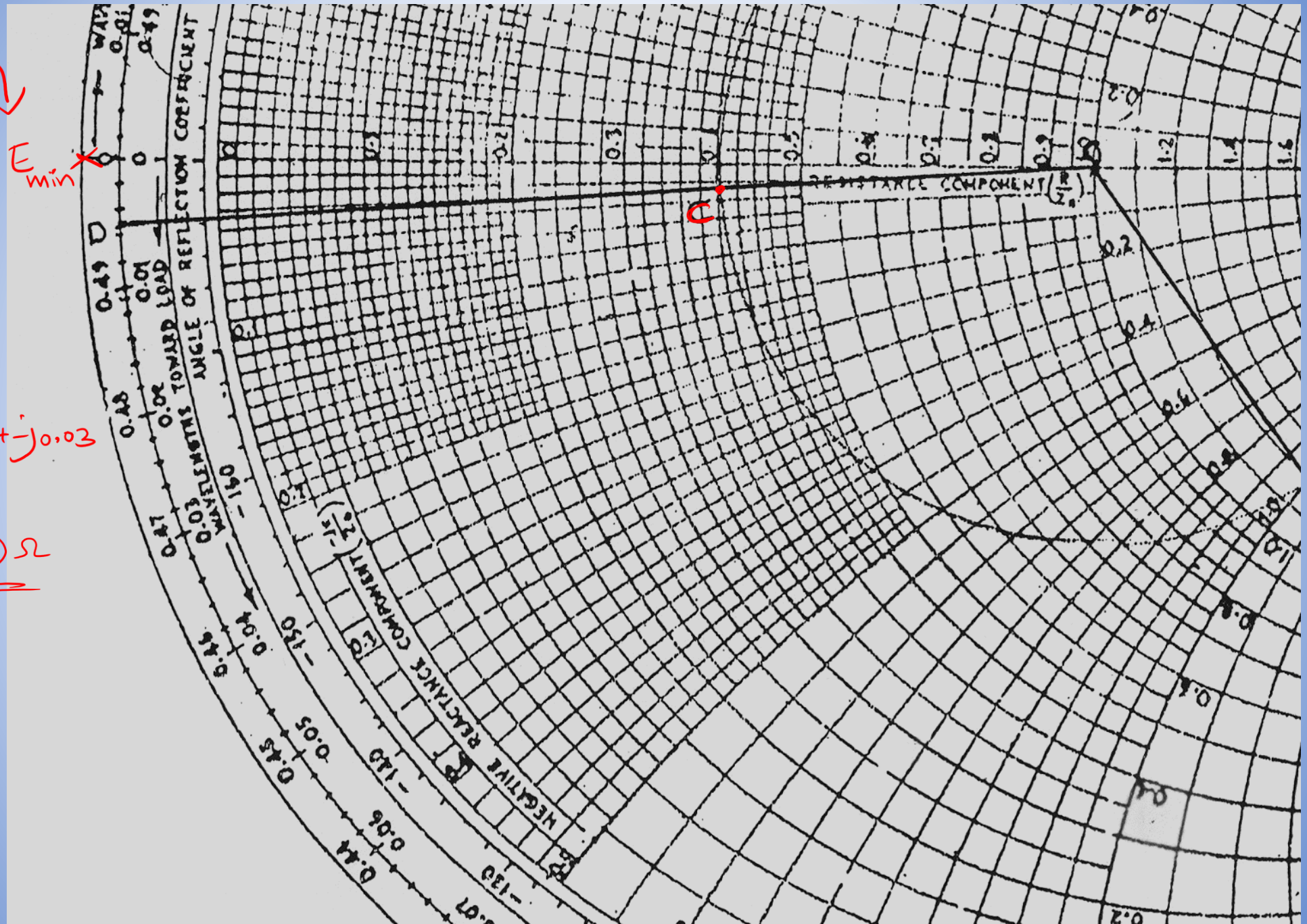
# Draw SWR circle with $r_a = 2.5$




Emin
0.175 λ
towards load
A
B
$A = \frac{Z_R}{Z_0}$
$= 1.2 - j1.0$
$\therefore Z_R = (120 - j100)\ \Omega$

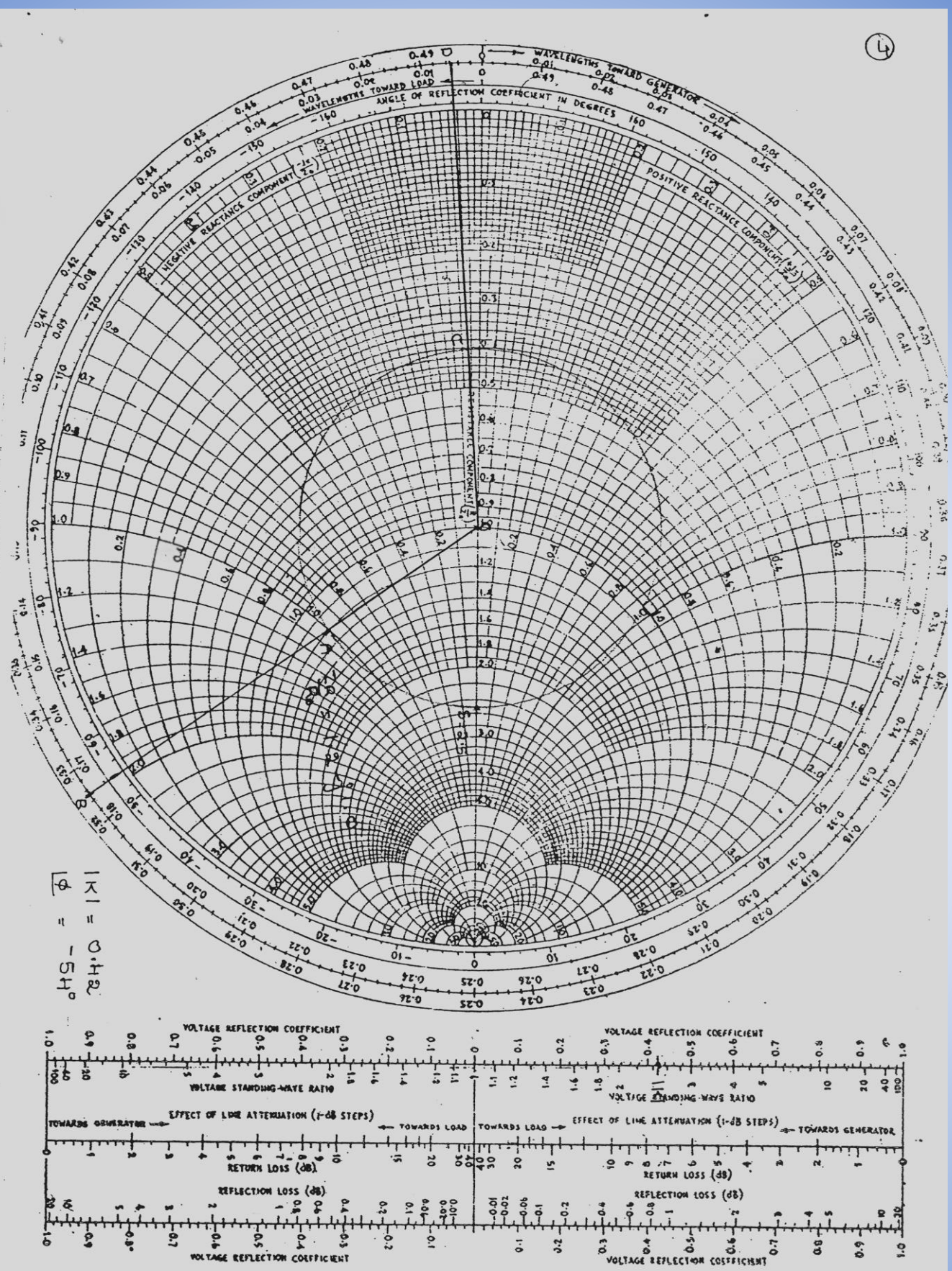

VOLTAGE REFLECTION COEFFICIENT
VOLTAGE STANDING-WAVE RATIO
EFFECT OF LINE ATTENUATION (1-dB STEPS)
TOWARDS GENERATOR
TOWARDS LOAD
RETURN LOSS (dB)
REFLECTION LOSS (dB)
VOLTAGE REFLECTION COEFFICIENT


CONDUCTANCE COMPONENT (G/Yo)
ANGLE OF TRANSMISSION COEFFICIENT IN DEGREES
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
RADIALLY SCALED PARAMETERS
TOWARD LOAD —>
CENTER
S =
|K|




$$c = \frac{Z_g}{R_0} = 0.4 - j0.03$$

$$Z_g = (40 - j3)\,\Omega$$

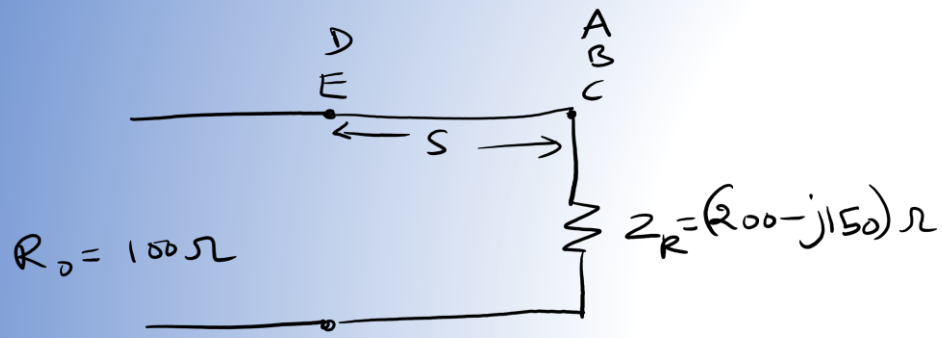



Smith Chart

SC/SBC

# Problem 7:

- Design a single stub matching network to match a load (200 – j150) Ω and $R_0$ = 100 Ω. The operating frequency is 30 MHz

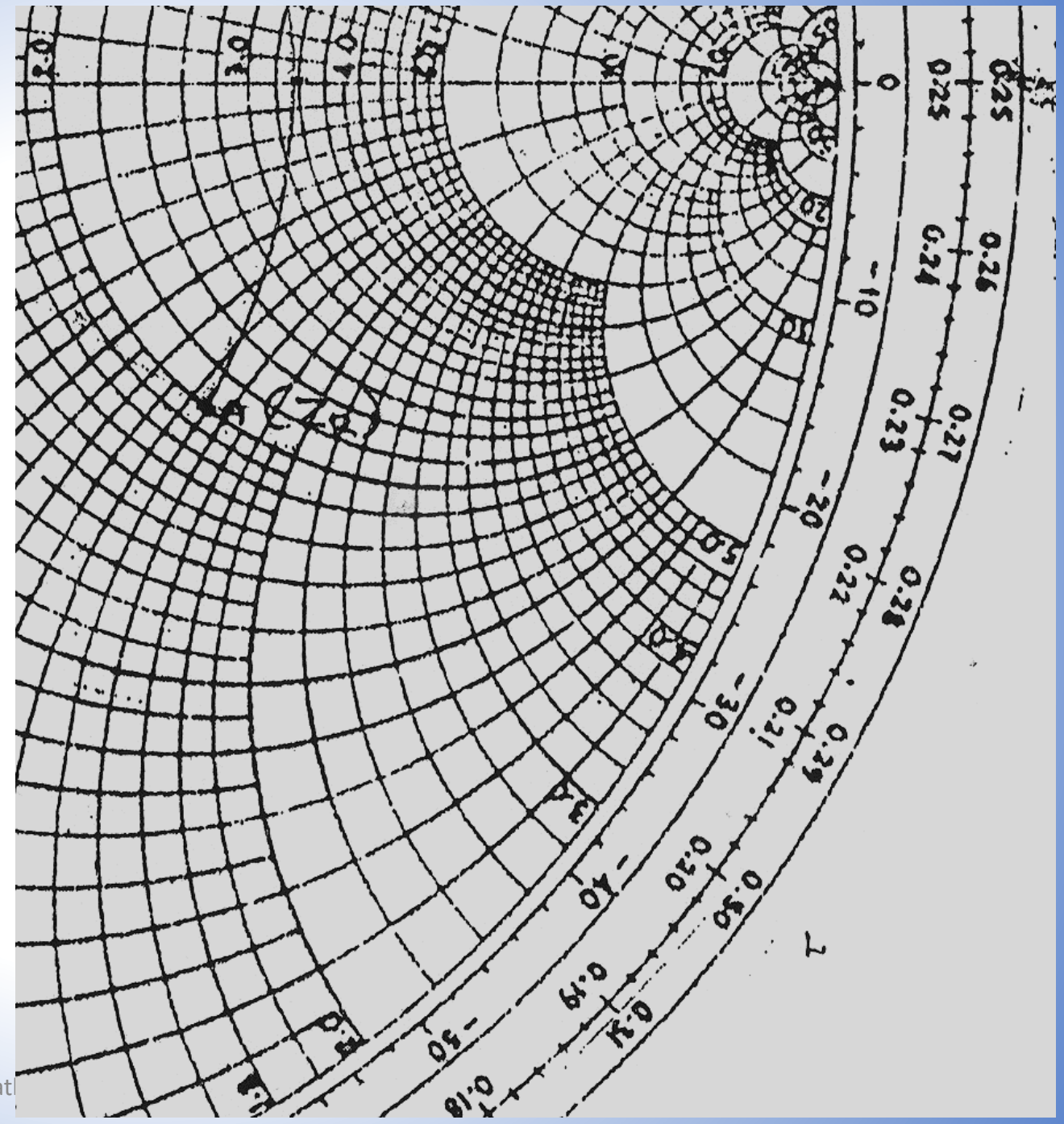



$$f = 30\,MHz \Rightarrow \lambda = \frac{c}{f} = 10\,m$$

$$\frac{Z_R}{R_0} = \frac{200 - j150}{100} = \underline{2 - j1.5\ (\text{Normalized})}$$

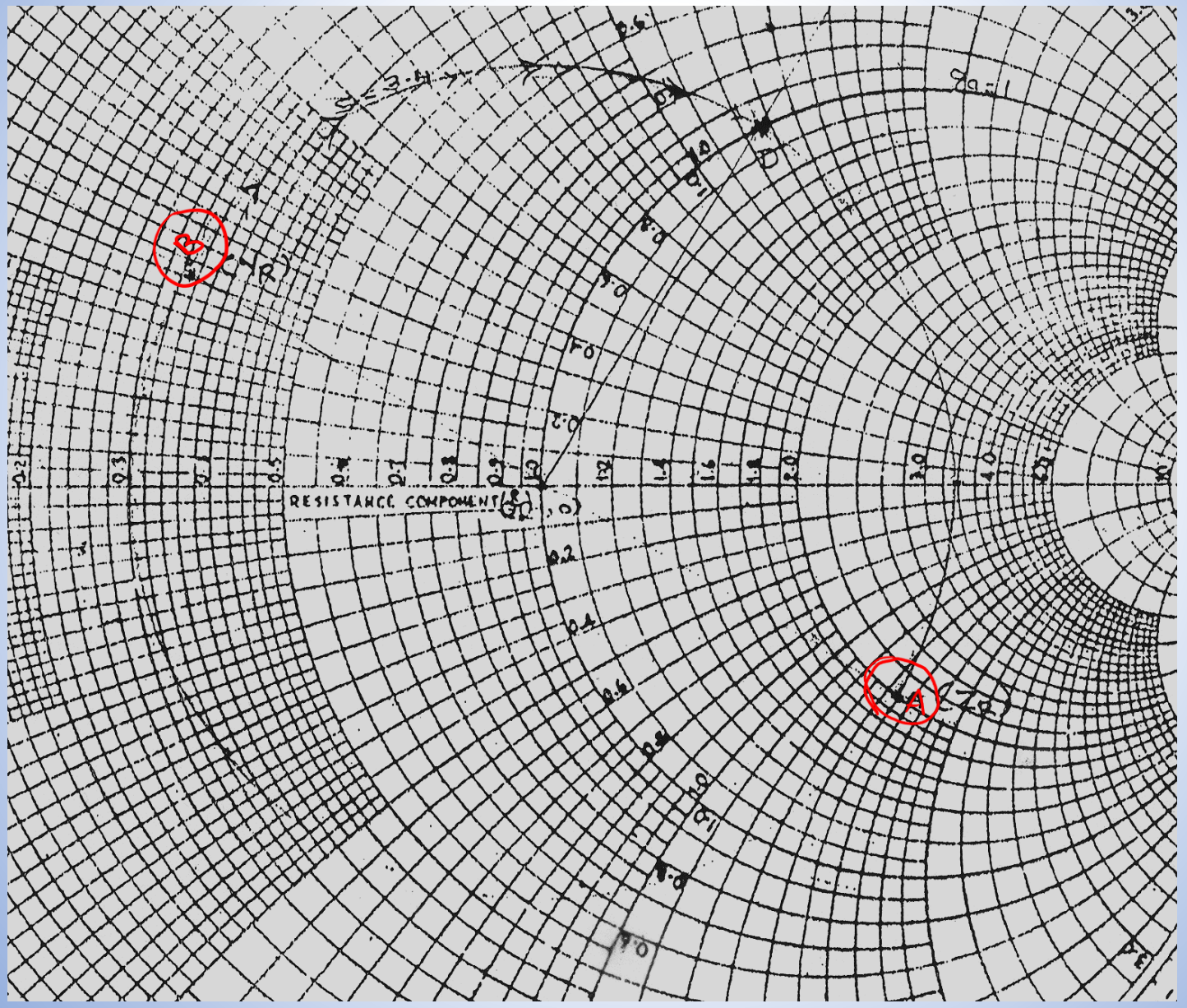
# Steps:

1. Normalize the load impedance (2-j1.5) and locate this point as A

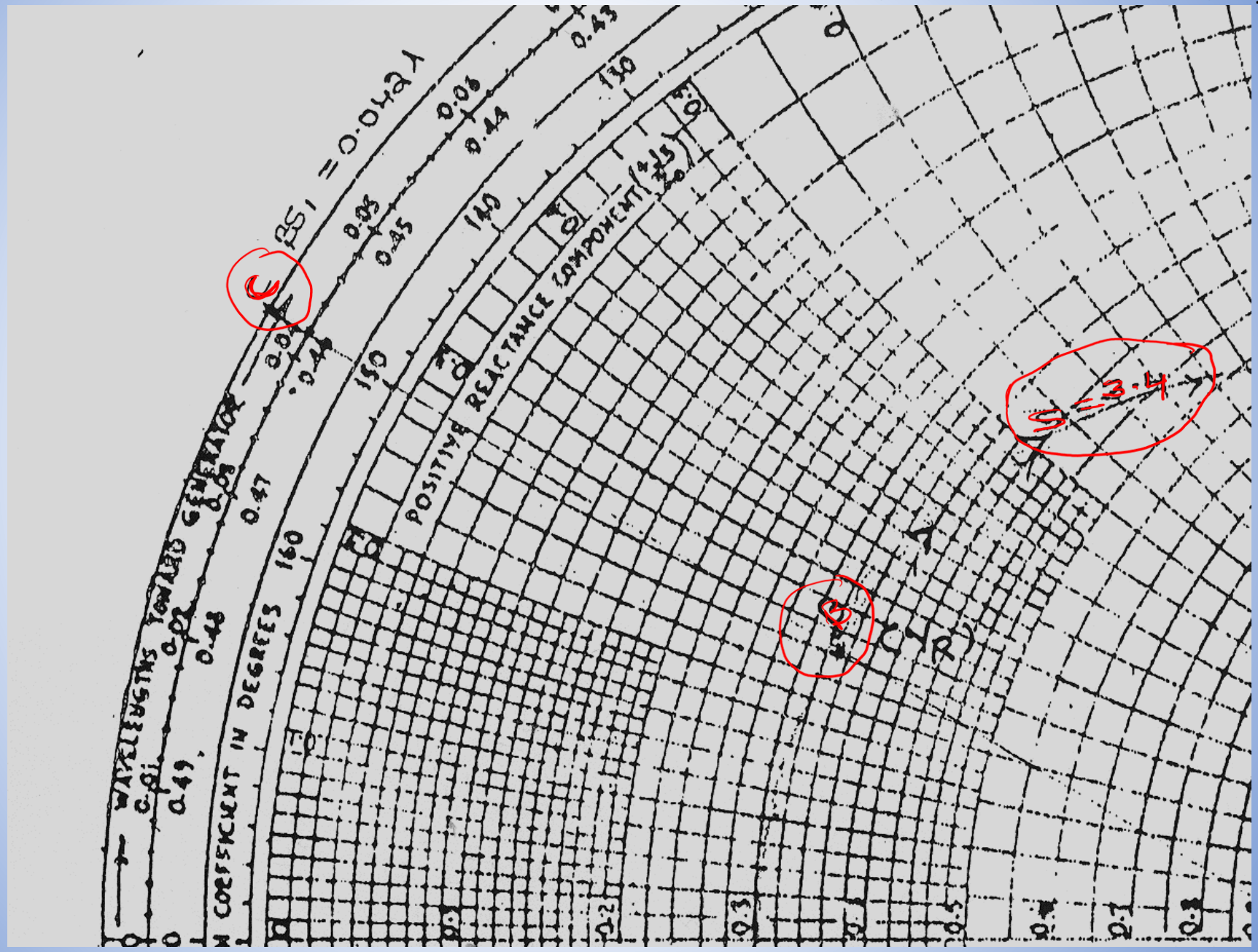


2. With $O_{(0,0)}$ as center and OA as radius draw constant S circle

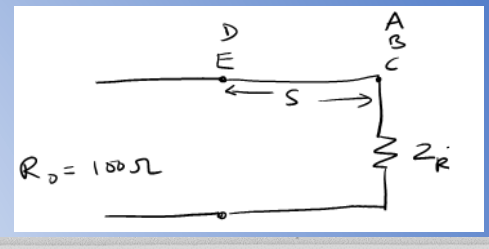
3. Locate diagonally opposite point B

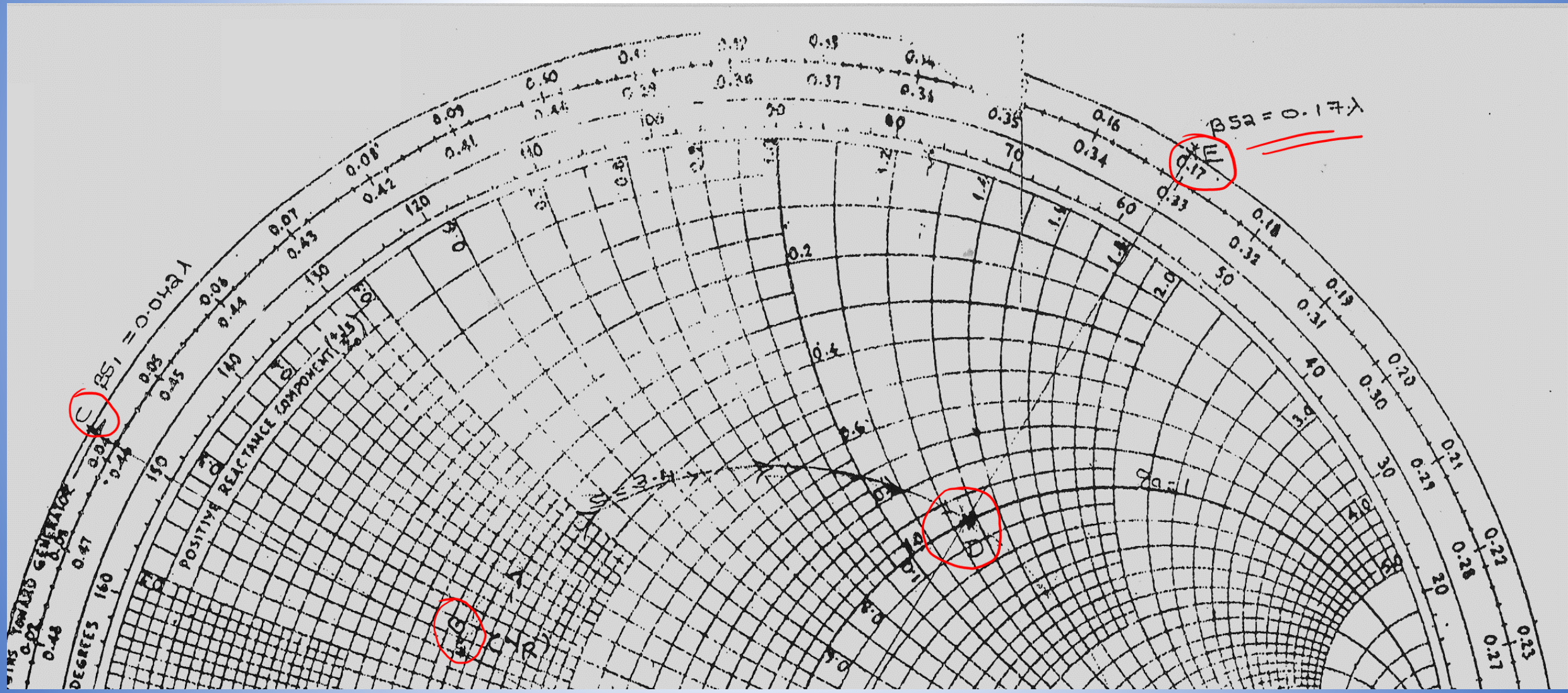


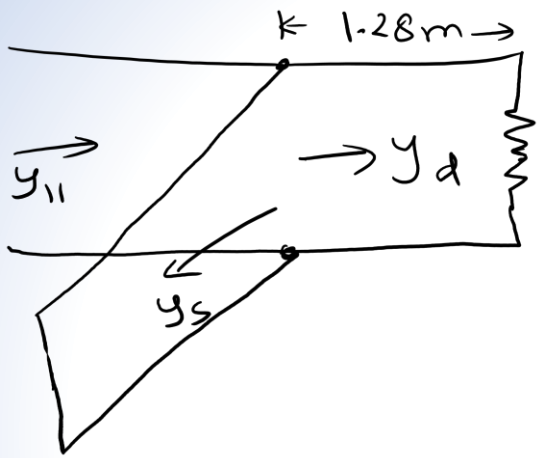
# 4. Locate B on constant S circle and C on constant βs circle. Read $\beta s_1$



5. From B move along SWR circle till it cuts $g_a = 1$ circle. Mark this point as D.
6. Extend it to get point E.
7. Position of stub = $\beta s2 - \beta s1 = 0.128\ \lambda = 1.28$ m
8. Read $Y_D = (1 + j1.3)$. The stub must provide a susceptance of (-j1.3)






$$S = \beta S_2 - \beta S_1$$

$$= 0.17\lambda - 0.042\lambda$$

$$= 0.128\lambda$$

Since $\lambda = 10m$, $S = \underline{1.28m}$

$$y_{11} = y_d \pm y_s = 1$$

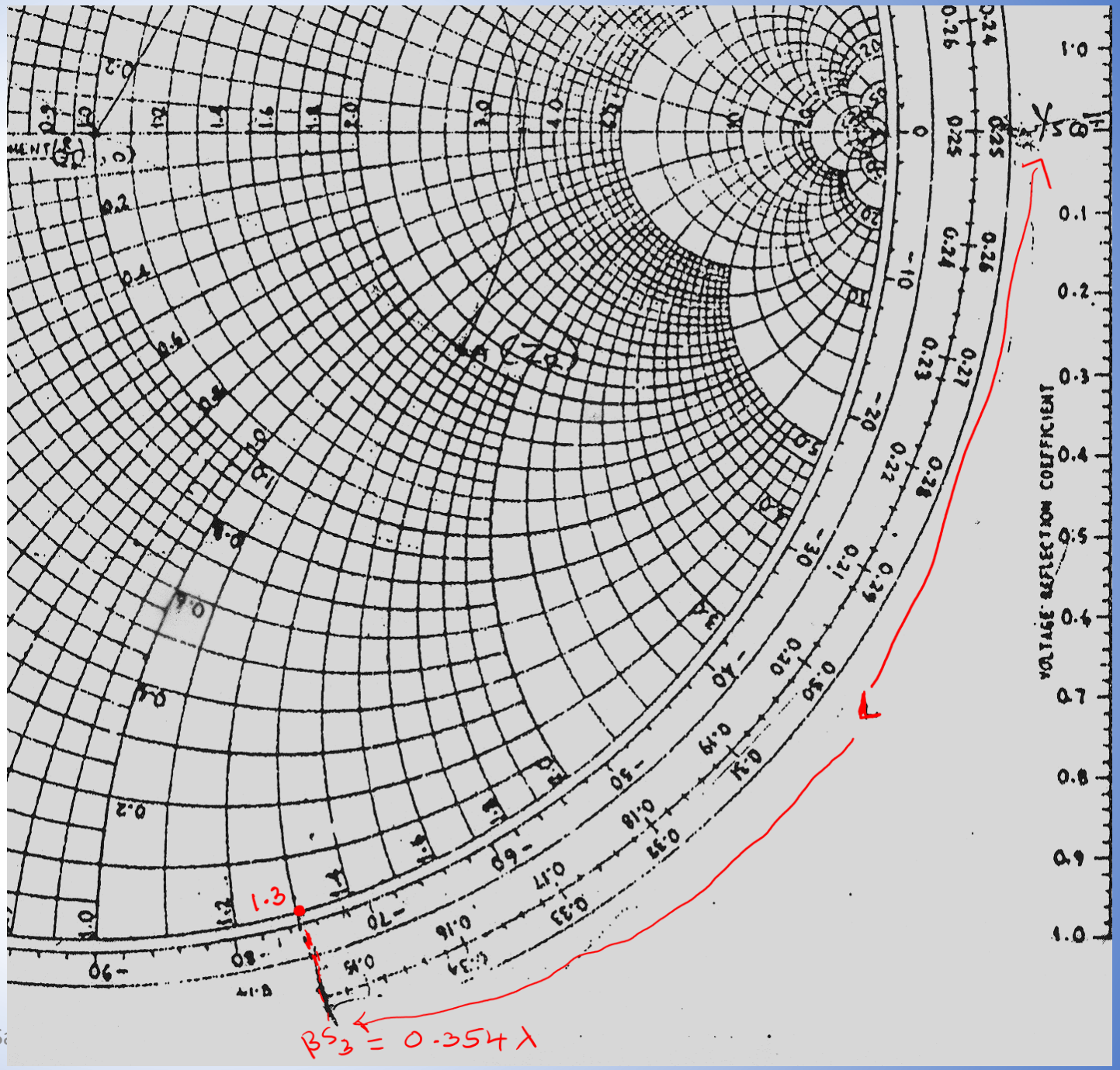
To find the value of susceptance:

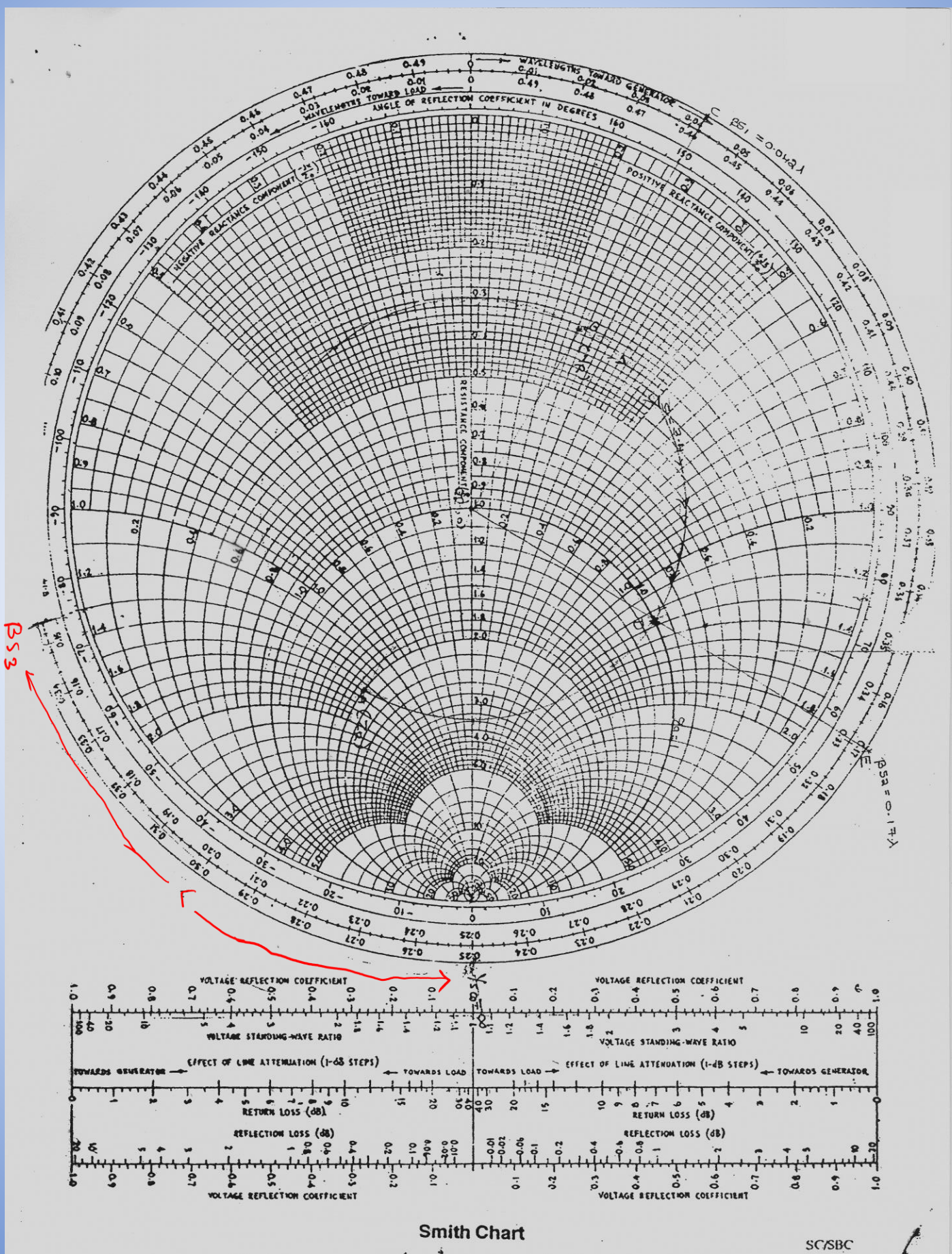
$$y_D = 1 + j1.3$$

We are suppose to design a stub which gives susceptance of $y_s = \underline{-j1.3}$

9. Identify (-j1.3) arc on the chart. Extend the arc to cut the βs circle. Read the value of $\beta s_3$

10. The short circuited stub must provide (-j1.3). Therefore length of the stub (Ysc = ∞) L = 0.354λ – 0.25λ = 0.104λ = <u>1.04m</u>






Smith Chart

SC/SBC

Problem 8: A lossless line of characteristic impedance $R_0 = 50\Omega$ is to be matched to a load $Z_l = 50 / [2+ j3.732]\ \Omega$ by means of a lossless short circuited stub. The characteristic impedance of stub is 100Ω. Find the stub position (closest to the load) and length so that match is obtained.

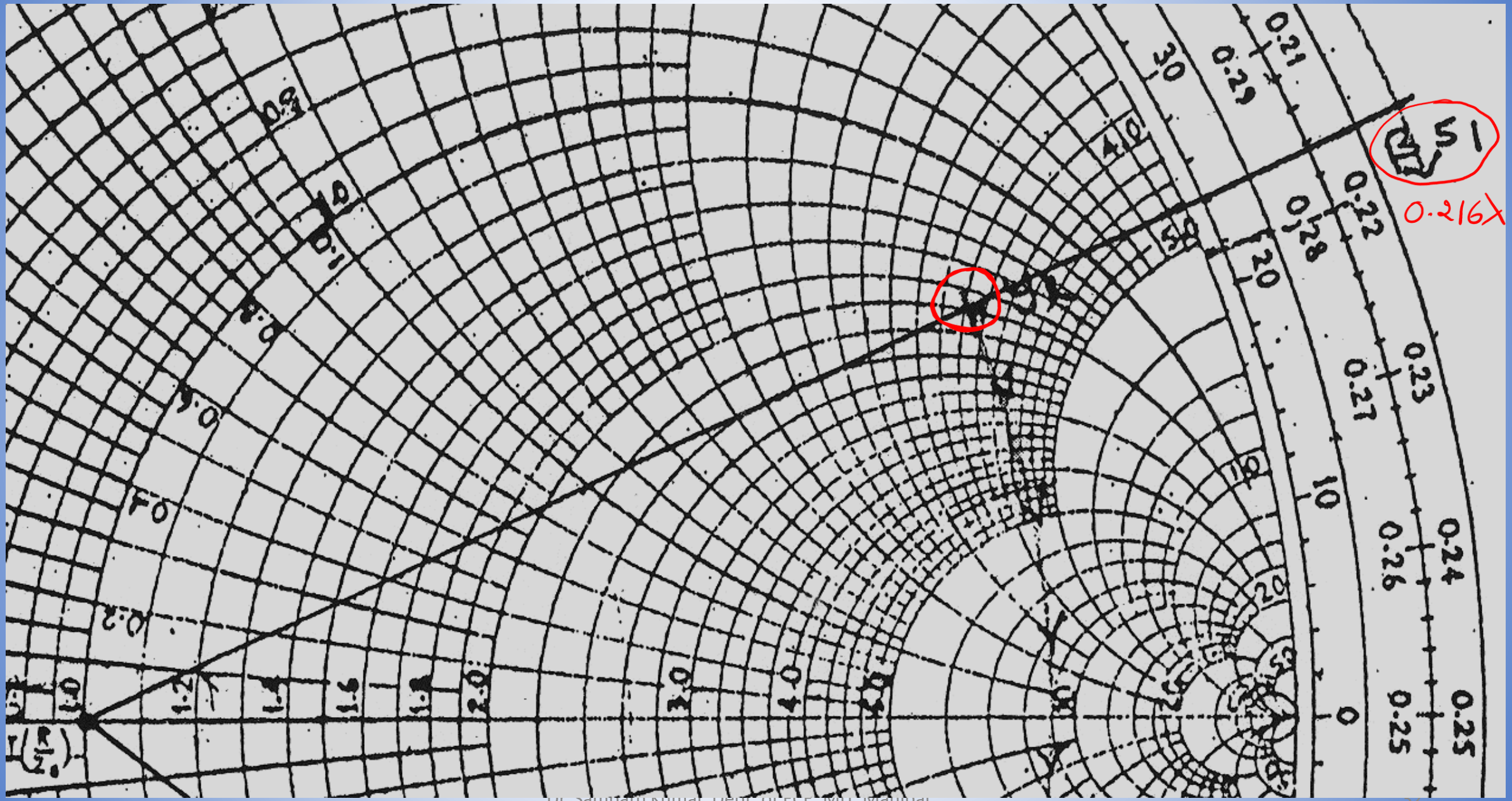
Steps:

1) Normalized impedance

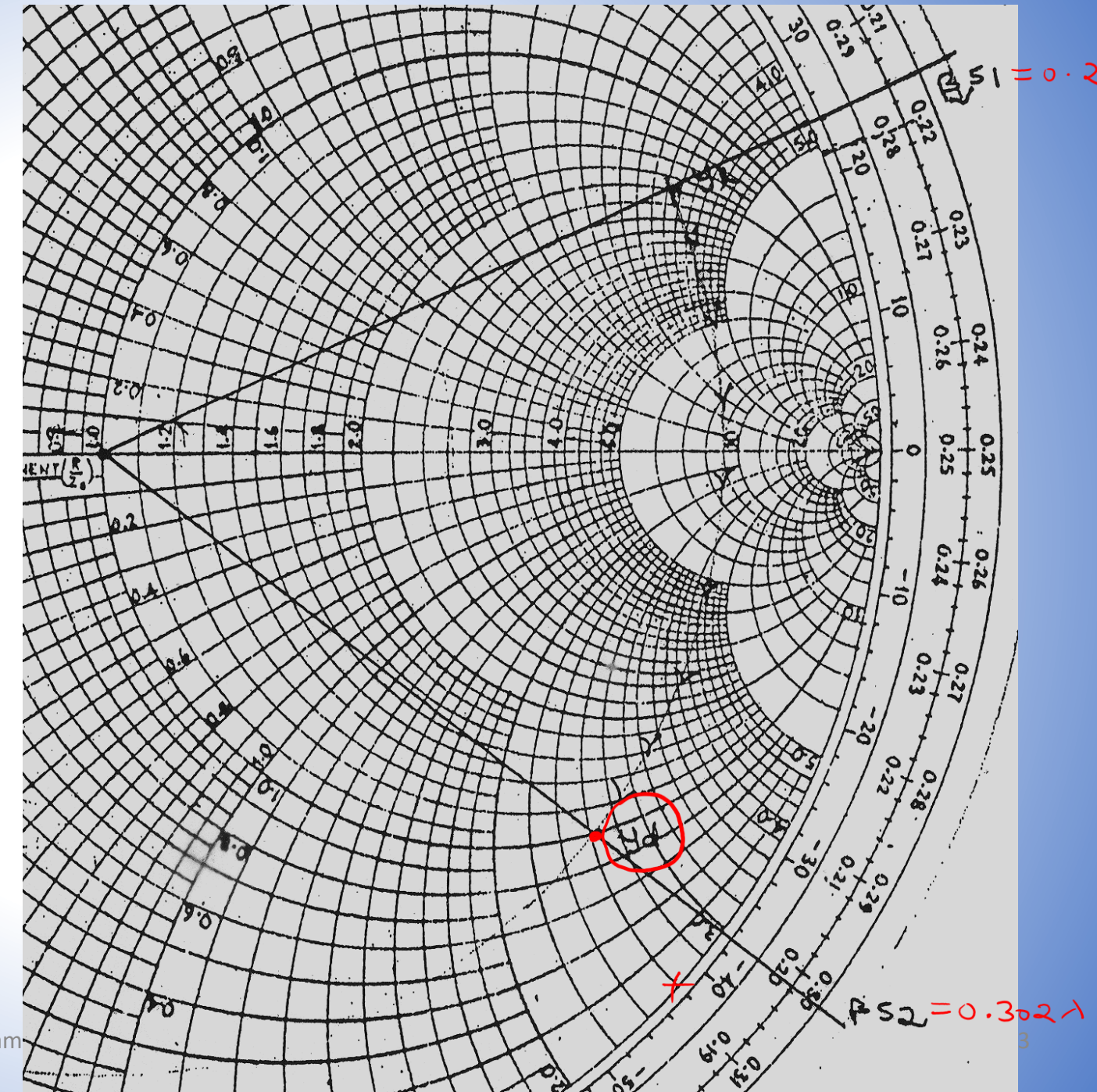
$$\frac{Z_l}{R_0} = \frac{1}{2+j3.732}$$

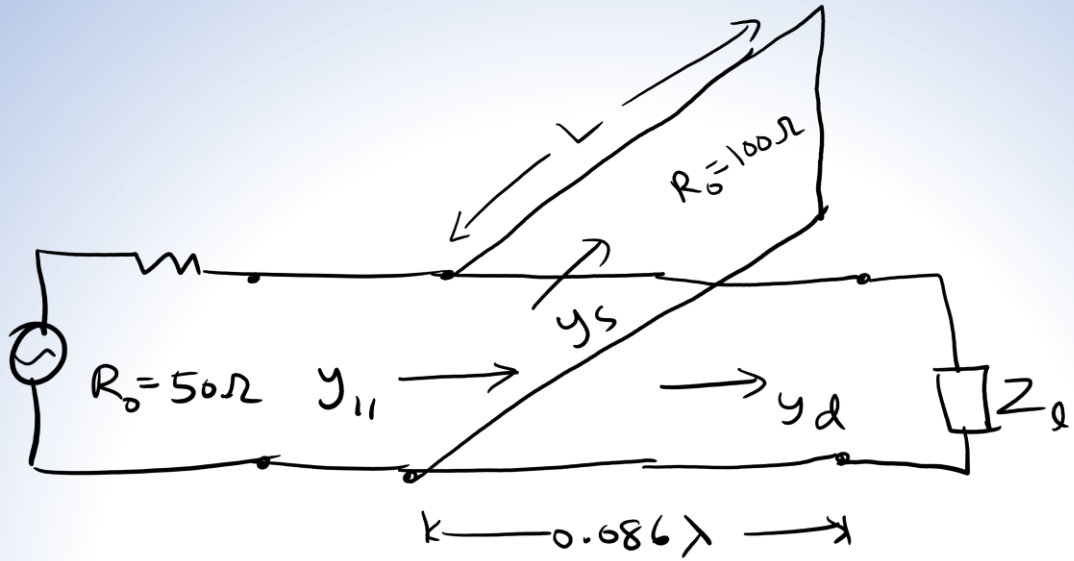
$\Rightarrow$ normalized admittance $= 2+j3.732$

2 + j3.732


0.216λ
0.09
0.1
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0.19
0.20
0.21
0.22
0.23
0.24
0.25
0.26
0.27
0.28
0.29
0.30
0.40
0.50
0.20
0.10
0
0.25
0.24
0.23
0.22
0.21
0.20
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00

- Move along SWR circle till it cuts $g_a$ = 1 circle to get βs2
- S = Bs2 - β1
  = (0.302 λ – 0.216 λ) = 0.086 λ
- Yd = 1-j2.6






$$Y_{11} = Y_d \pm Y_s$$

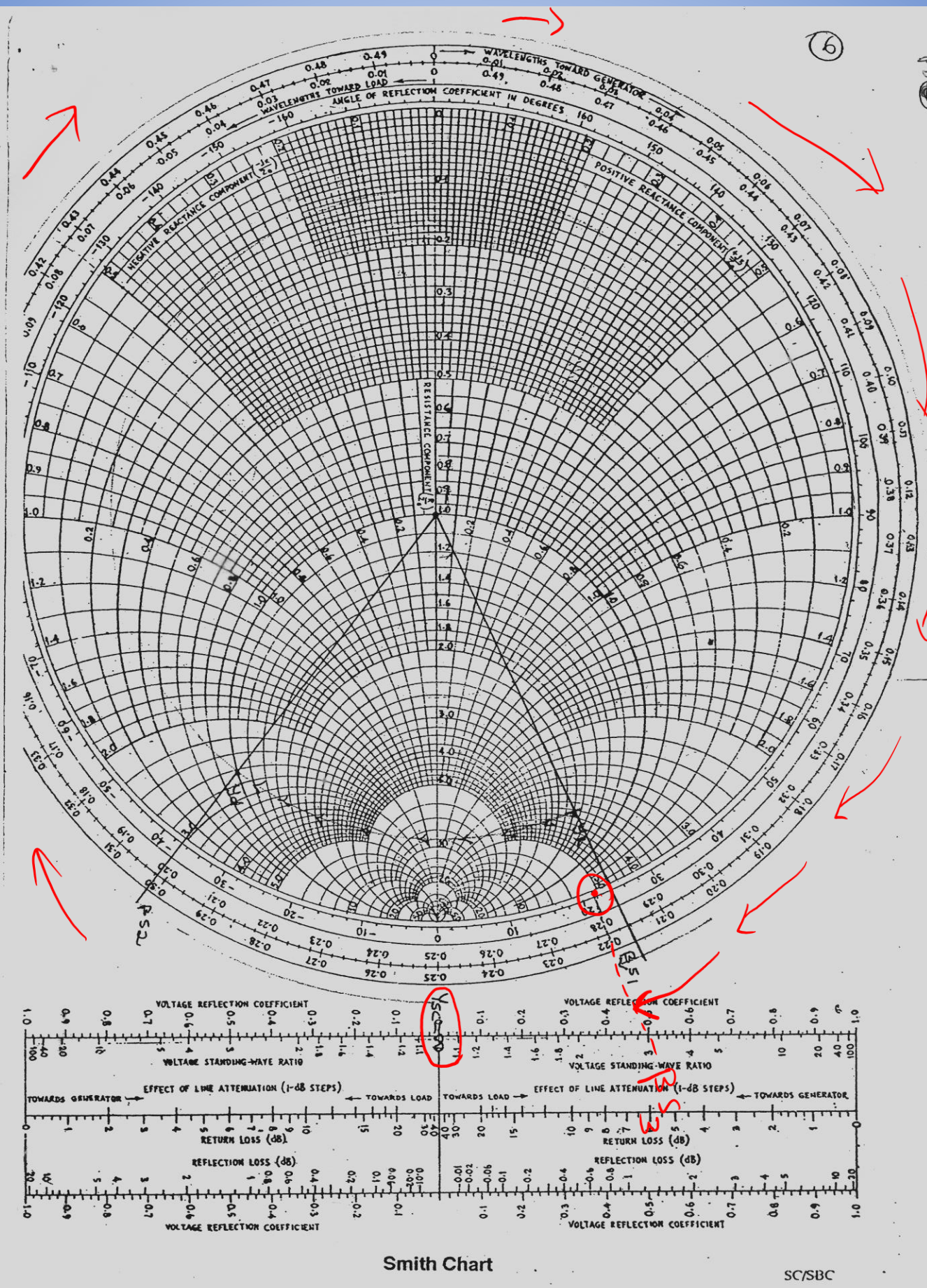
$$Y_{11} Y_0 = Y_d Y_0 \pm Y_s Y_{0s}$$

$$Y_s \cdot Y_{0s} = Y_{11} \cdot Y_0 - Y_d Y_0$$

$$Y_s = Y_{11} - Y_d \left(\frac{Y_0}{Y_{0s}}\right)$$

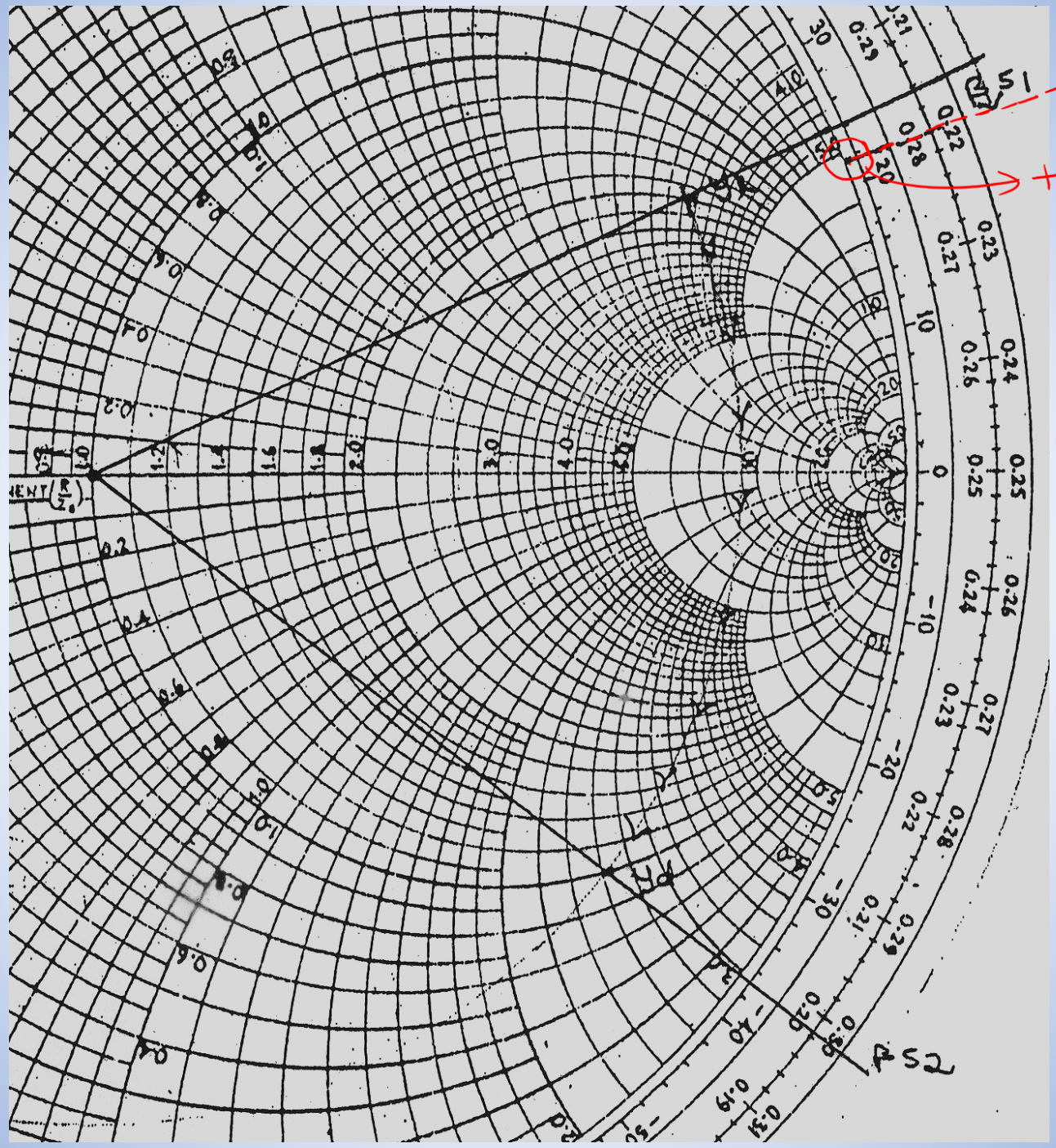
$$= 1 - (1 - j2.6)\frac{(Y_{50})}{\left(\frac{1}{100}\right)}$$

$$= j5.20$$




Smith Chart

SC/SBC


$\beta S_3 = 0.22\lambda$
$+j5.20$
$L = (0.25 + 0.22)\lambda$
$= 0.47\lambda$


Thank
you